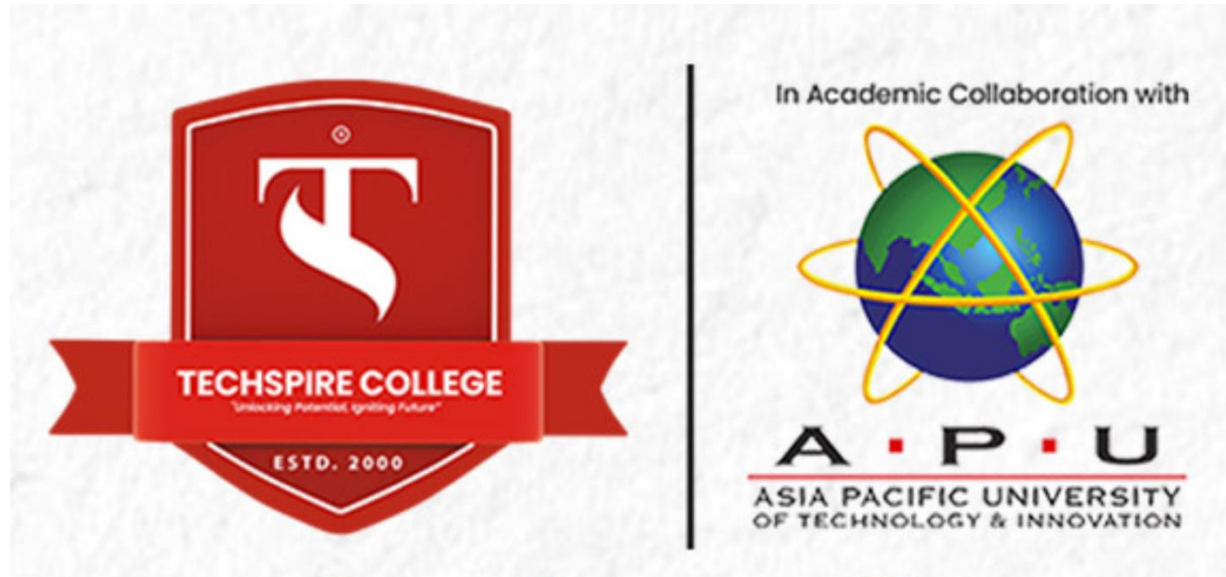




OOP in Java

Slide 2: Variables

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What Are Variables?

- Imagine a Box for Storing Information
 - Think of a variable as a box with a label. You can put information in the box and take it out whenever you need it. The label helps you remember what's inside.
 - Imagine you have a box labeled "name." You can put "John" inside the box. Whenever you need the name, you can look in the box.



Variable in Java

A variable is a place where we store data (like a number or a word) in our program so we can use it later.

Java has many types of variables, but for now, let's look at two simple ones:

- **String:** Stores words or text.
- **int:** Stores whole numbers (integers).

Creating a Variable in Java

- To create a variable, you first need to tell Java **what type of information** you want to store and then give it a **name**.
 - **String**: This means you're creating a box that stores words (text).
 - **name**: This is the label you give to the box.
 - **"Nimesh"**: This is what you're putting inside the box (the value).
 - **int**: This means you're creating a box that stores a whole number.
 - **age**: This is the label for the number box.
 - **25**: The number you're storing.

```
String name = "Nimesh"; // Stores the word "John"  
int age = 25;           // Stores the number 25
```

Using Variables to Print Information

- Once you've created a variable, you can use it in your program to display the information.
- **Explanation:**
 - "My name is ": This is just text.
 - + name: This combines (or **concatenates**) the text with the value stored in the name variable.
 - The same goes for age.

```
String name = "Nimesh"; // Stores the word "John"  
int age = 25;           // Stores the number 25  
  
System.out.println("My name is " + name);  
System.out.println("I am " + age + " years old");
```

```
My name is Nimesh  
I am 25 years old  
  
Process finished with exit code 0
```

Why Use Variables?



Variables help us store information that can **change** or that we need to use in different parts of the program. Instead of repeating the same information over and over again, we store it once in a variable and use it whenever we need to.



Example: Instead of writing "Nimesh" and 25 multiple times, we store them in name and age. If Nimesh's name changes, we only need to update it once in the variable, and it will update everywhere in the program.

Exercise



TASK 1: CREATE A VARIABLE TO STORE
YOUR NAME AND PRINT IT.



TASK 2: CREATE A VARIABLE TO STORE
YOUR FAVORITE NUMBER AND PRINT IT.

Summary



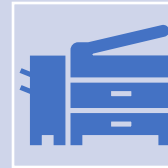
Variable: A storage box for information in your program.



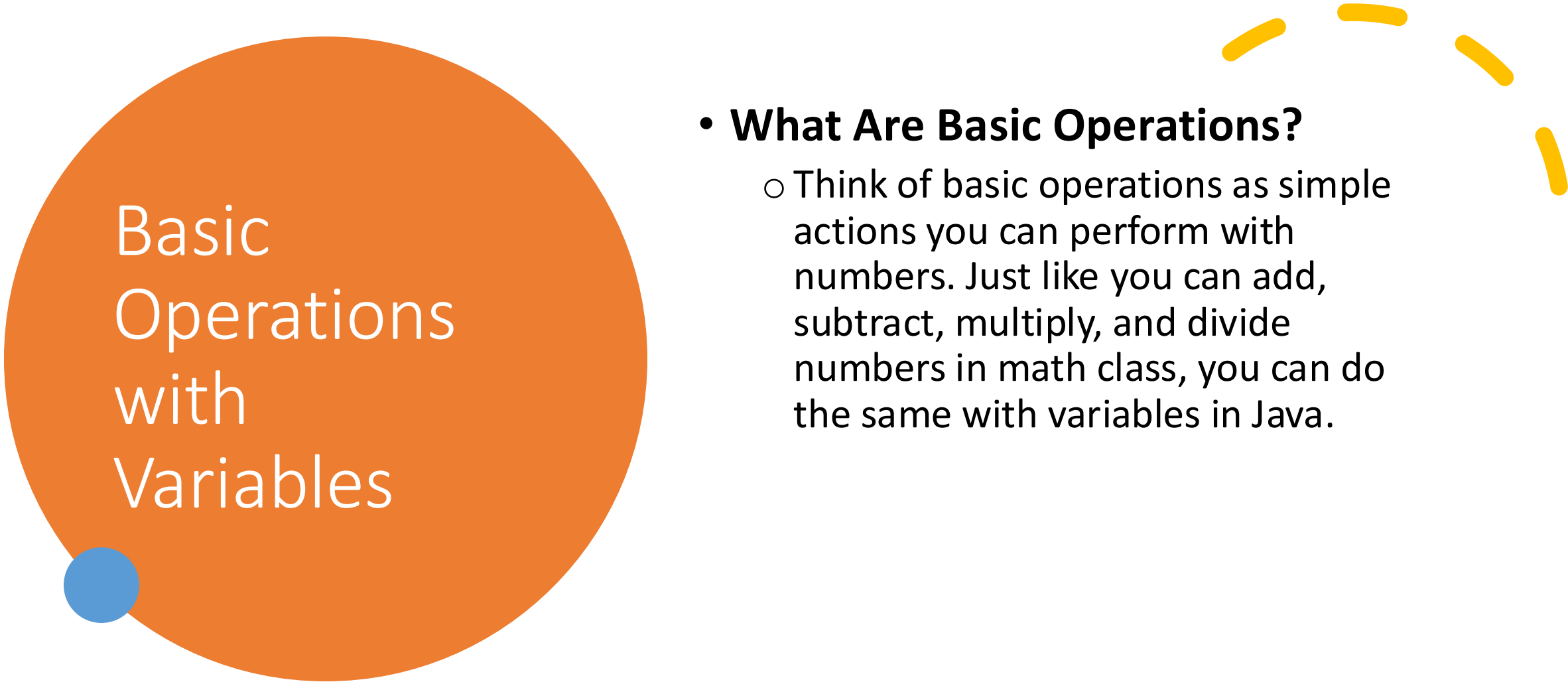
String: Stores text.



int: Stores whole numbers.



Using Variables: You can use variables to print information by combining them with text.



Basic Operations with Variables

- **What Are Basic Operations?**
 - Think of basic operations as simple actions you can perform with numbers. Just like you can add, subtract, multiply, and divide numbers in math class, you can do the same with variables in Java.

Arithmetic Operations

Addition (+): Adds two numbers together.

- `int sum = 5 + 3;`
- The variable `sum` will store the result of 5 plus 3, which is 8.

Subtraction (-): Subtracts one number from another.

- `int difference = 10 - 4;`
- The variable `difference` will store the result of 10 minus 4, which is 6.

Multiplication (*): Multiplies two numbers.

- `int product = 7 * 2;`
- The variable `product` will store the result of 7 times 2, which is 14.

Division (/): Divides one number by another.

- `int quotient = 20 / 4;`
- The variable `quotient` will store the result of 20 divided by 4, which is 5.

Modulo (%): Finds the remainder after division.

- `int remainder = 15 % 4;`
- The variable `remainder` will store the remainder when 15 is divided by 4, which is 3.

Using Variables in Operations

- **Combining Variables:** You can also perform operations using variables instead of just numbers.

```
int number1 = 8;  
int number2 = 5;
```

```
int additionResult = number1 + number2;  
int subtractionResult = number1 - number2;  
int multiplicationResult = number1 * number2;  
int divisionResult = number1 / number2;  
int moduloResult = number1 % number2;
```

```
System.out.println("Addition: " + additionResult);  
System.out.println("Subtraction: " + subtractionResult);  
System.out.println("Multiplication: " + multiplicationResult);  
System.out.println("Division: " + divisionResult);  
System.out.println("Modulo: " + moduloResult);
```

Addition: 13

Subtraction: 3

Multiplication: 40

Division: 1

Modulo: 3

Order of Operations

Just like in math, Java follows the order of operations (PEMDAS):

- 1. Parentheses

- 2. Exponents

- 3. Multiplication and Division (from left to right)

- 4. Addition and Subtraction (from left to right)

```
int result = (5 + 3) * 2;  
// Result will be 16 because (5 + 3) is 8, and 8 * 2 is 16.
```

Exercise

- **Task 1:** Create two variables with your favorite numbers and perform all basic operations with them.
- **Task 2:** Experiment with parentheses to change the order of operations.

```
int result1 = (6 + 2) * 3;  
int result2 = 6 + (2 * 3);  
System.out.println("Result  
1: " + result1); // Expect  
24  
System.out.println("Result  
2: " + result2); // Expect  
12
```

Summary



Basic Operations: Addition, subtraction, multiplication, division, and modulo.



Using Variables: Perform operations with variables to store and manipulate results.



Order of Operations: Follow PEMDAS to ensure correct results in calculations.